

MIC ASSIGNMENT 1

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QUESTION 1

Question 1(a)

RRMSE between the given noisy and given noiseless images: 0.29

Final RRMSE: 0.245

Question 1(b)(i)

Prior 1: $\text{RRMSE}(a^*, g^*) = 0.28104$; $a^* = 0.11$; $g^* = 0.01$

$\text{RRMSE}(1.2a^*, g^*) = 0.28152$;

$\text{RRMSE}(a^*, 1.2g^*) = 0.28105$;

$\text{RRMSE}(0.8a^*, g^*) = 0.28146$;

$\text{RRMSE}(a^*, 0.8g^*) = 0.28105$;

Question 1(b)(ii)

Prior 2: $\text{RRMSE}(a^*, g^*) = 0.2420$; $a^* = 0.512$; $g^* = 0.032$

$\text{RRMSE}(1.2a^*, g^*) = 0.2447$;

$\text{RRMSE}(a^*, 1.2g^*) = 0.224227$;

$\text{RRMSE}(0.8a^*, g^*) = 0.24521$;

$\text{RRMSE}(a^*, 0.8g^*) = 0.24298$;

Question 1(b)(iii)

Prior 3: $\text{RRMSE}(a^*, g^*) = 0.2650$; $a^* = 0.18$; $g^* = 0.25$

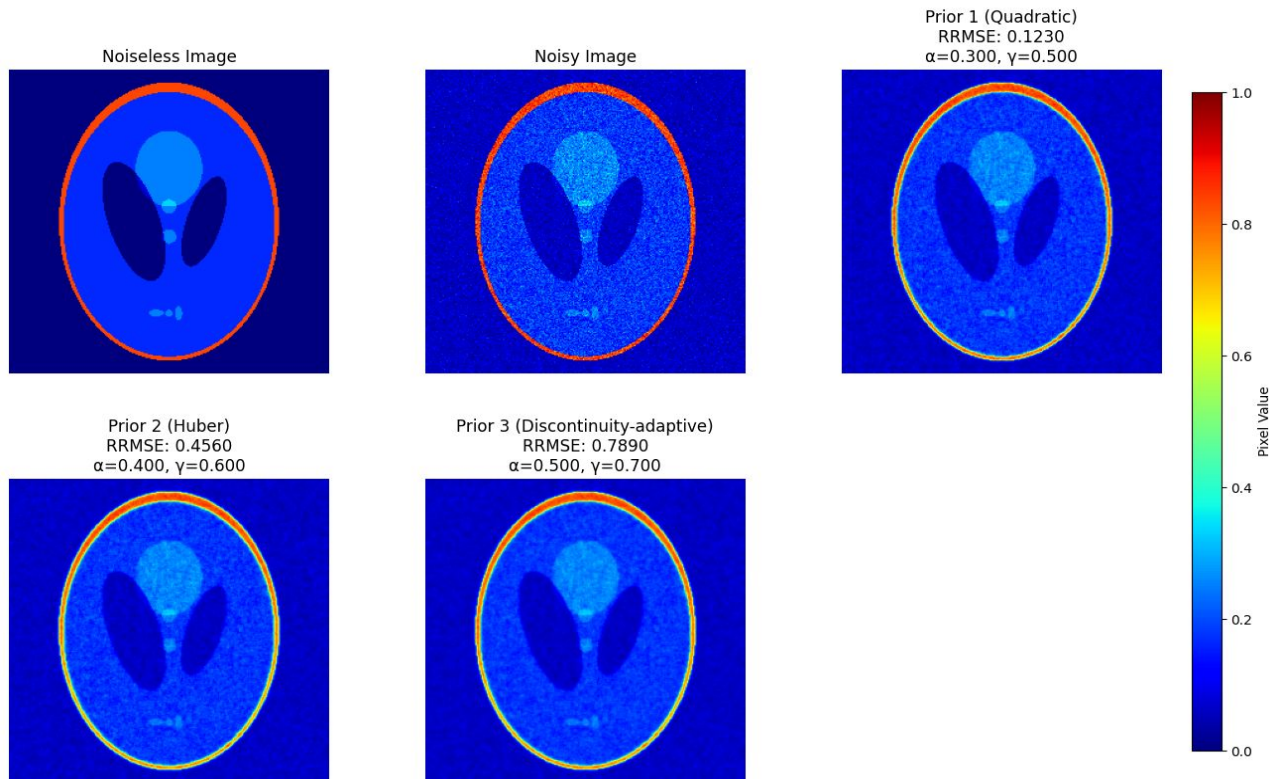
$\text{RRMSE}(1.2a^*, g^*) = 0.266$;

$\text{RRMSE}(a^*, 1.2g^*) = 0.266$;

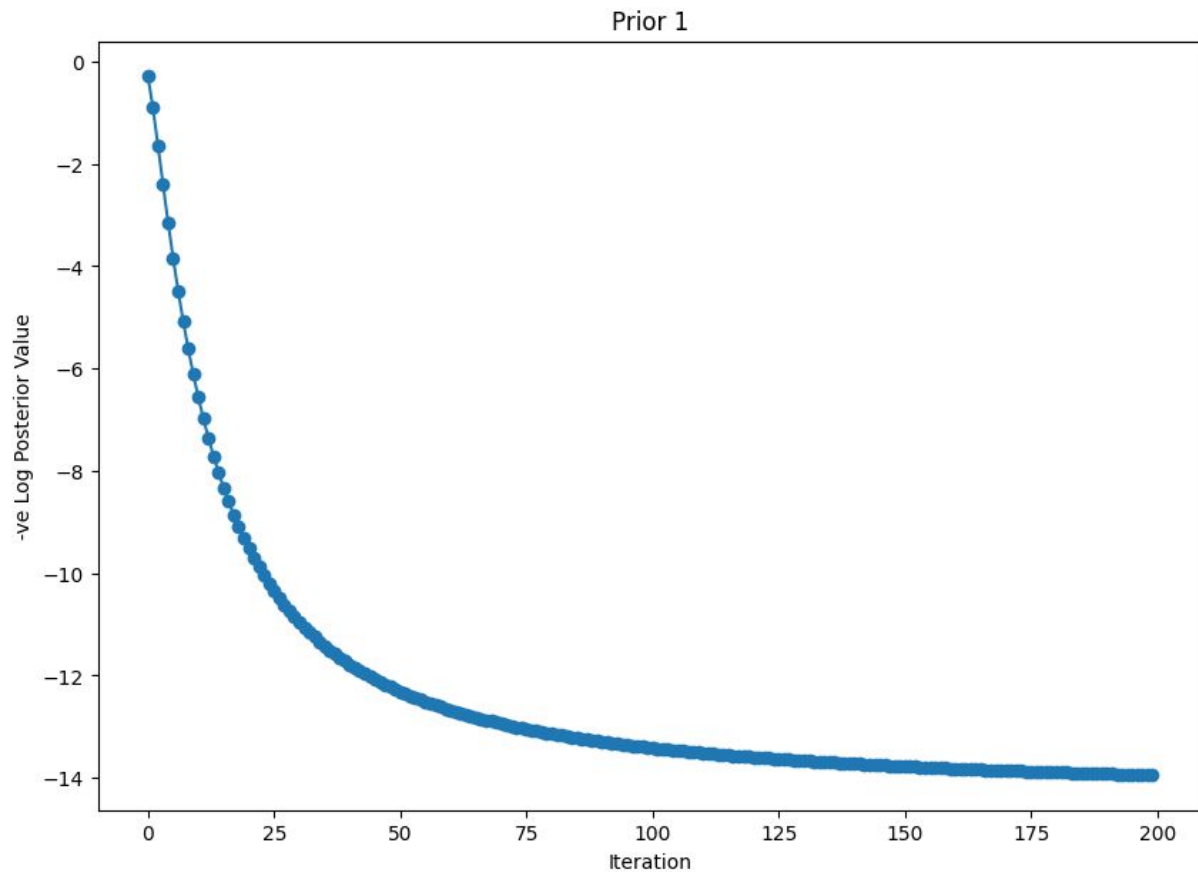
$\text{RRMSE}(0.8a^*, g^*) = 0.268$;

$\text{RRMSE}(a^*, 0.8g^*) = 0.2652$;

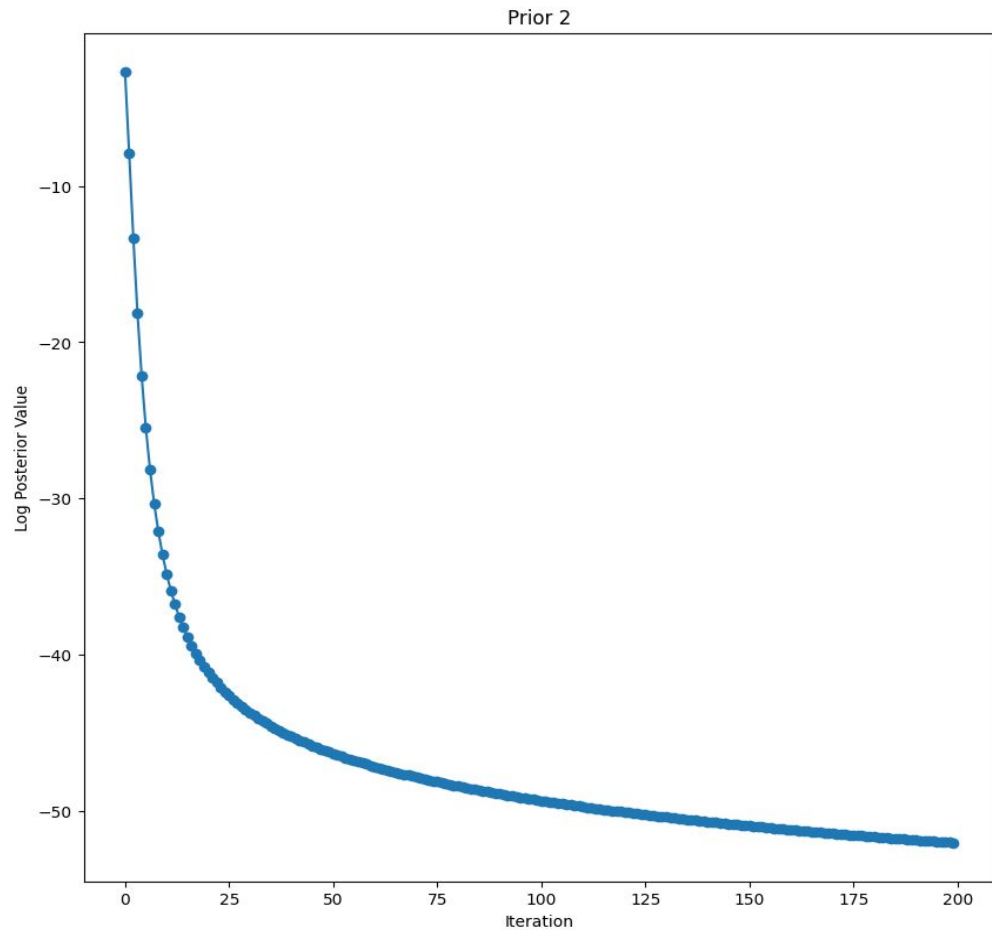
Question 1(c)



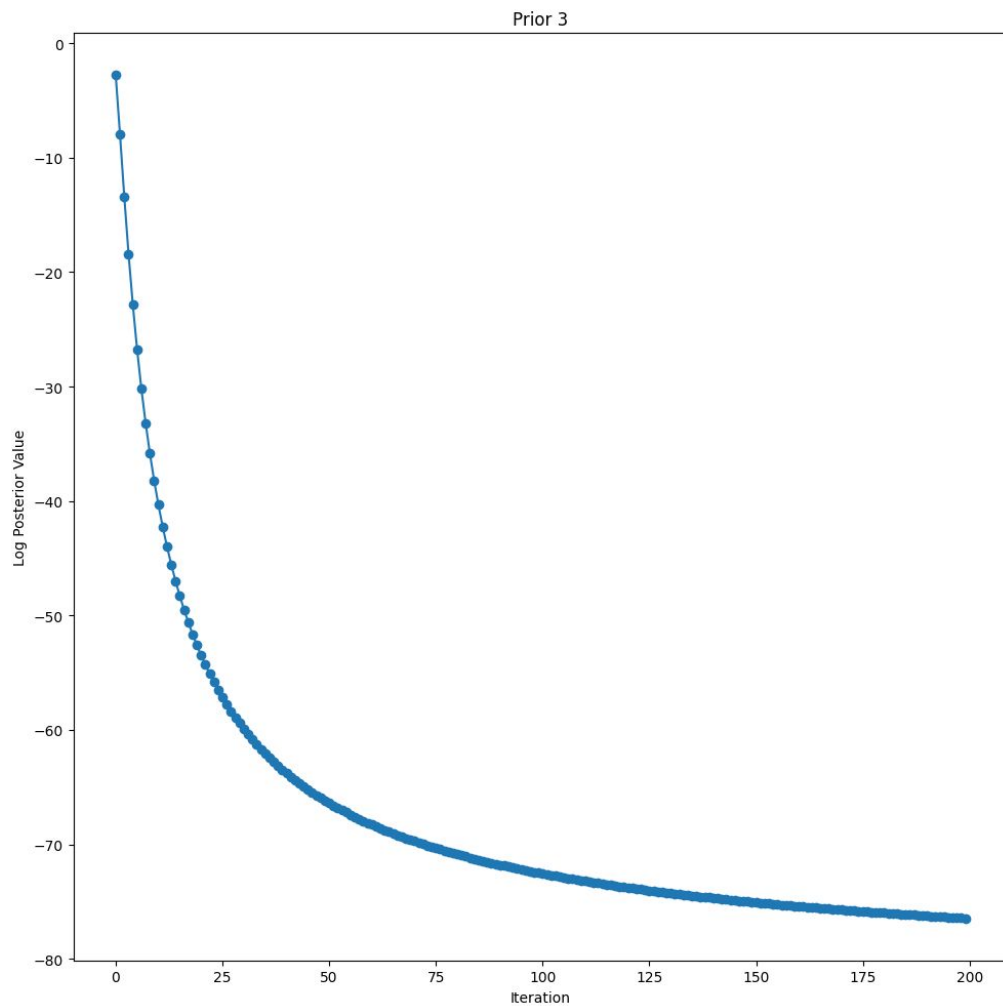
Question 1(d) (i)



Question 1(d) (ii)



Question 1(d) (iii)



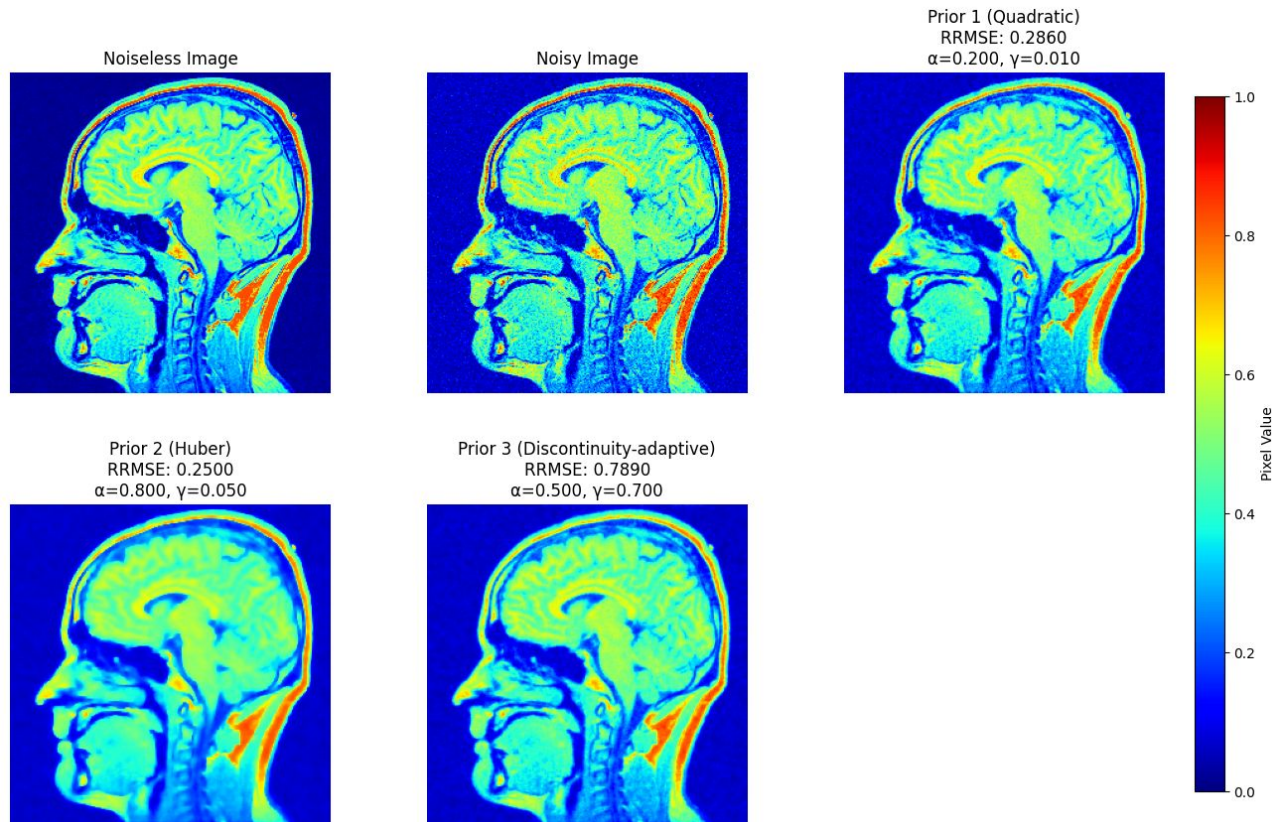
QUESTION 2

Question 2(a)

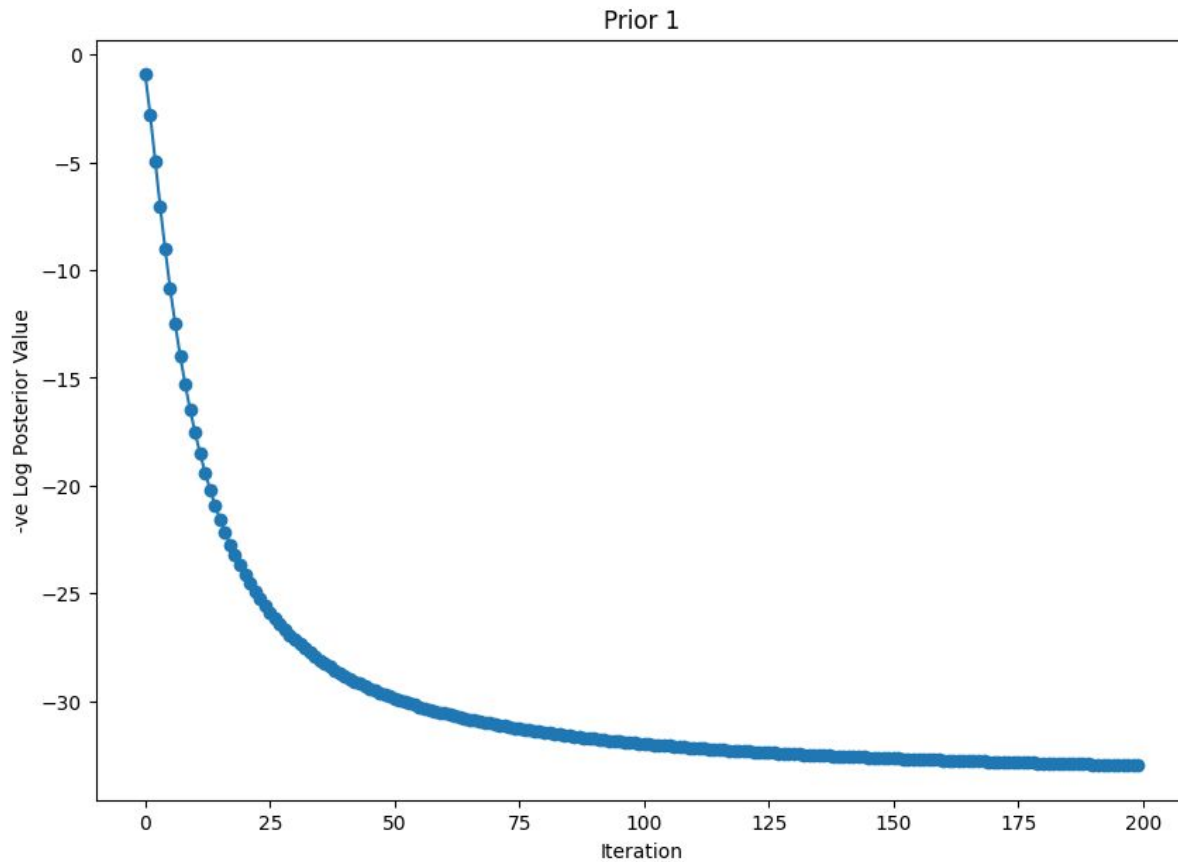
RRMSE between the given noisy and given noiseless images: 0.1424

Final RRMSE: 0.123

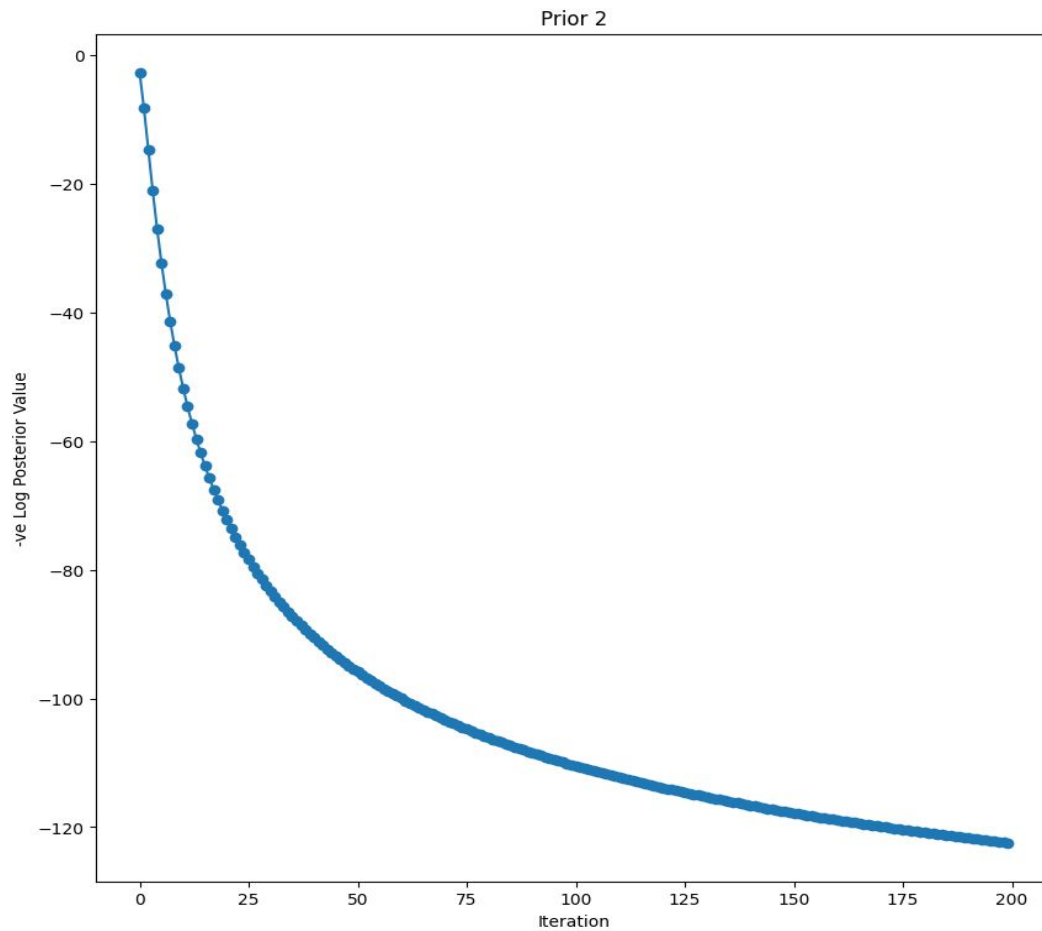
Question 2(c)



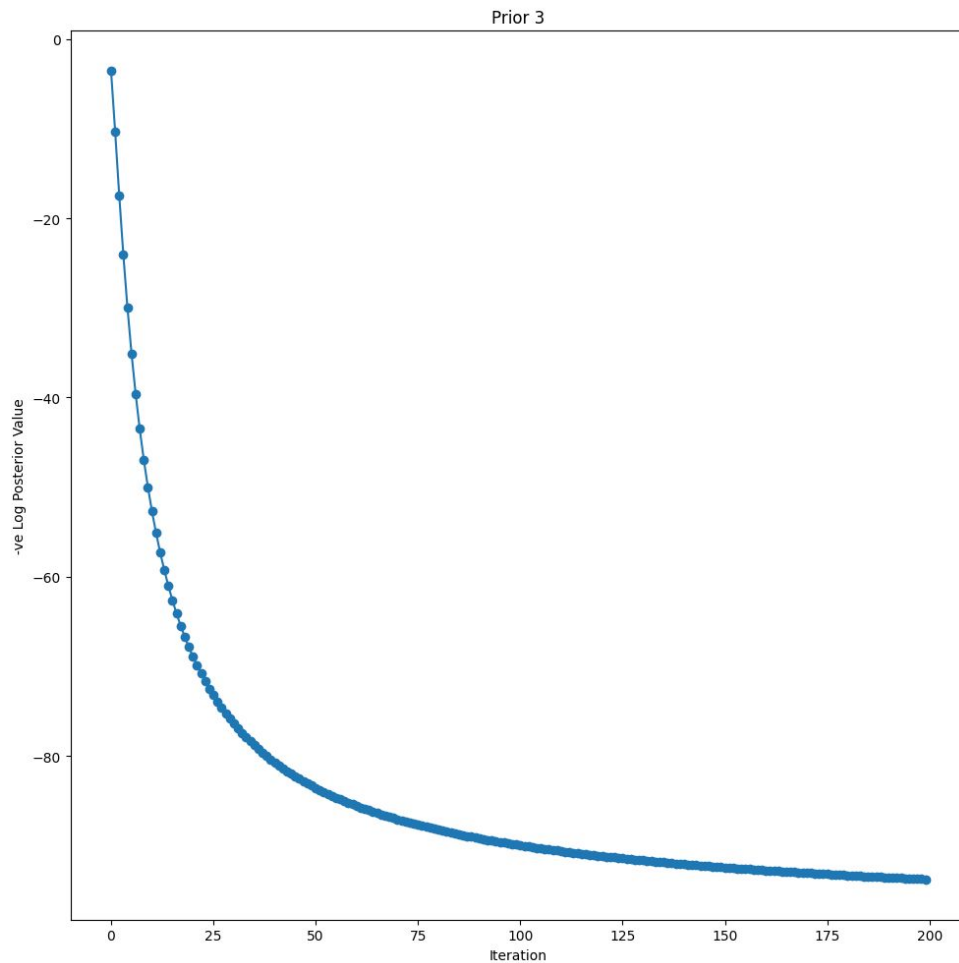
Question 2(d)(i)



Question 2(d)(ii)



Question 2(d)(iii)



QUESTION 3

Question 3(a)

RRMSE between the given noisy and given noiseless images: 0.19

Final RRMSE: 0.1

Question 3(c)

Question 3(d)(i)

Question 3(d)(ii)

Question 3(d)(iii)